

Edge Detection Analysis

Canny vs. Marr–Hildreth Comparison

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CHAPTER 1

Introduction and Motivation

Edge detection plays a pivotal role in image analysis, forming the backbone of many computer vision applications such as segmentation, object recognition, and scene understanding.

This report presents an in-depth comparison between two classical methods:

- **Canny Edge Detector**[\[1\]](#) – a gradient-based technique emphasizing optimal detection, localization, and minimal response.
- **Marr–Hildreth (Laplacian of Gaussian)**[\[2\]](#) – a zero-crossing based method derived from the theory of visual perception.

Both methods were implemented from scratch, refined through several algorithmic improvements, and evaluated quantitatively over 200 test images with reference ground truths.

The aim is to examine not only their numerical performance (Precision, Recall, F1-score) but also their qualitative edge structures and robustness under noise.

CHAPTER 2

Methodology Overview

Each method was implemented independently, with consistent pre-processing and evaluation pipelines to ensure comparability. Beyond textbook definitions, both algorithms were enhanced to improve noise resistance, edge sharpness, and adaptability to varying image contrast.

2.1 Common Pre-processing

Before applying either detector, each image underwent a custom noise-adaptive smoothing process:

- **Grayscale Conversion** Converts coloured image to Grayscale (0-255 intensity).
- **Gaussian Smoothing** ($\sigma = 3.0$, kernel size 11×11) removes high-frequency noise.
- **Adaptive Local Variance Weighting** suppresses flat areas more strongly than textured regions.
- **Normalization** scales output intensity between $[0,1]$ for stable gradient or LoG computation.

This preprocessing significantly stabilizes both detectors when handling noisy or low-contrast inputs.

CHAPTER 3

Canny Edge Detection

3.1 Basic Steps (Classical Algorithm)

The Canny detector follows a structured multi-stage approach:

1. Gaussian smoothing to remove noise.
2. Gradient computation using Sobel filters.
3. Calculation of gradient magnitude and direction.
4. Non-maximum suppression (NMS) to thin edges.
5. Double thresholding and hysteresis to finalize connected edges.

3.2 Improvements Added

The implementation extends the standard algorithm through several key refinements aimed at improving precision and reducing edge thickness:

- **Custom NMS threshold** (`res * (mag[i,j] > 0.08)`): Discards faint gradients during suppression, minimizing thick edges.
- **Adaptive Double Thresholding**: Uses percentile-based high thresholds (`np.percentile(Z, high_percentile)`) instead of fixed constants, automatically adapting to contrast variations across images.
- **Hysteresis Refinement**: Requires neighboring strong edges and magnitudes above $1.2\times$ the mean, reducing weak edge bridging.
- **Post-gradient Filtering**: Residual weak edges are suppressed by re-masking with strong gradients (`res * (mag > percentile)`).

- **Integration with Pre-processing:** The smoothed input significantly reduces salt-and-pepper noise and false edges.

Together, these improvements enhance both precision and visual clarity, resulting in thinner, more reliable contours compared to the standard OpenCV implementation.

CHAPTER 4

Marr–Hildreth (Laplacian of Gaussian) Edge Detection

4.1 Basic Steps (Classical Algorithm)

The Marr–Hildreth detector relies on finding zero-crossings in the Laplacian of Gaussian (LoG) response:

1. Apply Gaussian smoothing to the image.
2. Convolve with a Laplacian of Gaussian kernel.
3. Detect zero-crossings in the filtered image (where the sign of the Laplacian changes).
4. Mark those crossings as edges.

4.2 Improvements Added

The implemented version introduces several refinements for better edge localization and noise control:

- **Pre-smoothing step** ($\sigma = 3.0$): Removes minor noise frequencies before LoG computation.
- **Gradient magnitude constraint** ($grad_mag[i, j] > grad_thresh$): Combines LoG with Sobel gradients to eliminate false zero-crossings in flat regions.
- **Magnitude difference test** ($diff > mag_thresh$): Ensures only zero-crossings with significant contrast are retained.
- **Tunable parameters** ($\sigma, mag_thresh, grad_thresh$): Allow optimization for different image types.

- **Normalization of LoG response:** Ensures consistent response scales across varying resolutions.

These refinements transform Marr–Hildreth from a blob-prone operator into a precise contour detector capable of matching gradient-based algorithms in clarity.

CHAPTER 5

Experiments and Evaluation

5.1 Dataset and Setup

- **Images:** 200 noisy test images located in the `images/` folder.
- **Ground Truth:** Corresponding boundary maps (.mat files) from the BSDS dataset, containing multi-annotator averaged edges.
- **Evaluation Tolerance:** A 2-pixel neighborhood was allowed to handle minor misalignments between prediction and ground truth.

5.2 Evaluation Metrics

Each edge map was compared to its reference map using standard binary classification metrics:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad \text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

A binary dilation tolerance was applied during evaluation to accommodate small localization errors without penalizing accurate but slightly shifted detections.

CHAPTER 6

Experimental Results

6.1 Example Visual Comparisons



Figure 6.1: Visual comparison of three examples across both edge detectors. Each row shows one image: original, ground truth, Canny result, and Marr-Hildreth result.

6.2 Result Analysis

The overall evaluation scores across the 200 test images are summarized below. Each value represents the mean $\pm$ standard deviation across all examples. These metrics quantify the trade-off between precision (edge correctness) and recall (edge completeness).

Method	Precision	Recall	F1-Score
Canny	0.388 ± 0.152	0.542 ± 0.126	0.432 ± 0.116
Marr-Hildreth	0.350 ± 0.140	0.748 ± 0.097	0.457 ± 0.130

The Marr-Hildreth detector achieved higher **recall**, indicating that it detected a larger proportion of true edges. This comes at the cost of slightly lower **precision**, suggesting an increase in false detections. Overall, its F1-score of **0.457** surpasses Canny's **0.432**, confirming an improved balance between edge completeness and accuracy. Visual inspection further supports these findings: both Canny and Marr-Hildreth produced denser edge maps compared to the human-annotated ground truth, which typically highlights only the most prominent object boundaries. In contrast, the detectors captured finer structures such as facial features, textures, and background elements, offering richer geometric detail.

Between the two methods, Marr-Hildreth consistently produced **thicker and smoother edges**, a result of its Laplacian-of-Gaussian filtering, which blurs the image before zero-crossing detection. Canny's edges, meanwhile, remained thinner and more sharply localized due to its non-maximum suppression and hysteresis thresholding. Overall, both detectors generated high-quality edge maps, with Canny favoring precision and clarity, while Marr-Hildreth prioritized edge continuity and completeness. The complete evaluation results, including per-image metrics, were saved in `edge_metrics_comparison.csv` for further analysis.

Bibliography

- [1] John F. Canny. A computational approach to edge detection. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, PAMI-8(6):679–698, 1986.
- [2] David Marr and Ellen C. Hildreth. Theory of edge detection. *Proceedings of the Royal Society of London. Series B, Biological Sciences*, 207(1167):187–217, 1980.